

Kunal Pratap Singh

+91 6266235370 | kunalps6266@gmail.com | Rewa, IN 486001 | LinkedIn | GitHub | Portfolio | LeetCode

EDUCATION

VIT Bhopal University

B.Tech in Computer Science; CGPA: 8.39

Bhopal, India

2023 – 2027

SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

AI/ML & GenAI: PyTorch, Scikit-learn, OpenCV, Transformers, NLP, LangChain, Deep Learning, Feature Engineering, Model Evaluation, Data Analysis & Visualization, Gemini API, RAG

Backend & Cloud: FastAPI, Flask, MongoDB, Docker, AWS

Frontend: React, Next.js

Tools & Core: Pandas, NumPy, Git

INTERNSHIP

Indian Space Lab

Intern

Remote

June 2025 – July 2025

- Analysed drone flight telemetry data using Python (Pandas, NumPy); identified 3 key performance bottlenecks in autonomous navigation algorithms and modelled optimisation strategies for battery life and flight stability.
- Produced a technical report on algorithmic performance evaluation of navigation systems; findings used to guide development priorities.

PROJECTS

AI Interview Coach

Live | Code

Next.js 14, TypeScript, Node.js, Express, MongoDB, Gemini API

- Engineered an LLM-powered interview simulation platform with role-specific scenarios and real-time AI evaluation of technical accuracy, code quality, efficiency, and response quality.
- Implemented RAG-based feedback grounding using semantically similar coding problems and MongoDB retrieval to provide context-aware evaluations and actionable improvement suggestions.
- Built persistent interview history and performance analytics with MongoDB, alongside JWT and Google OAuth authentication, enabling personalized progress tracking across practice sessions.

DDoS Attack Detection & Mitigation in SDN using DNN

Demo | Code

Python, PyTorch, Ryu SDN, Mininet, FastAPI

- Built a real-time SDN-based DDoS detection pipeline using a deep neural network trained on InSDN flow data, engineering features from OpenFlow statistics including packet/byte counts, rates, flow duration, protocol fields, and inter-arrival metrics.
- Integrated ML inference with a Ryu OpenFlow 1.3 controller and Mininet testbed; polled switch flow statistics every 2 seconds, computed flow deltas, and classified traffic using an attack-probability threshold of 0.8.
- Automated attack mitigation by installing high-priority OpenFlow drop rules for detected source IPs and logging events; achieved 99.65% detection accuracy using BatchNormalization, Dropout, class balancing, Adam optimization, and early stopping.

RBI Regulatory Digest Agent

Demo | Code

LangChain, Gemini API, FastAPI, Agents, RAG

- Architected an agentic pipeline using LangChain and the Gemini API within the Hermes Agent framework to monitor, extract, and summarise RBI regulatory circulars; a live run against production RBI feeds processed 20 circulars into 46 action points across 8 departments, flagging 11 critical items with imminent deadlines.
- Published as a technical write-up on DEV.to as part of the Hermes Agent Challenge; the work earned a guest-post invitation from CoderLegion.